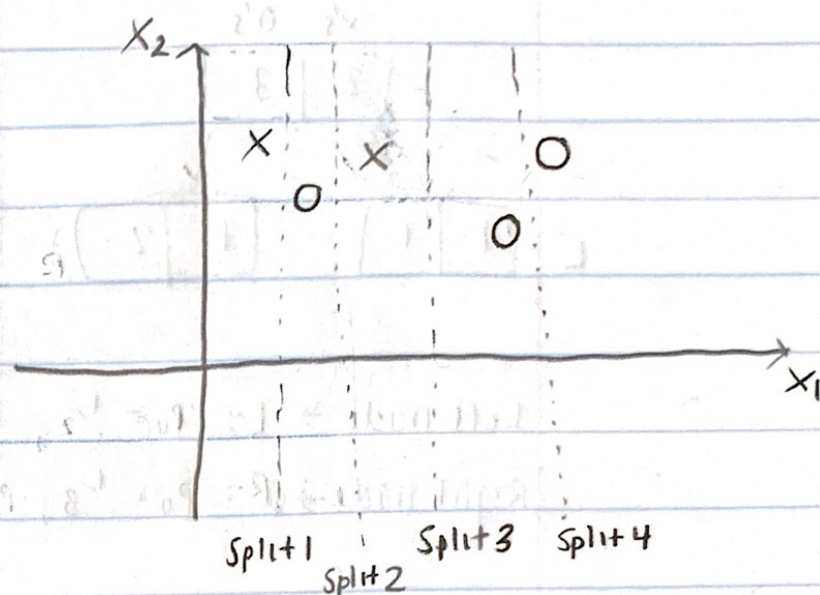
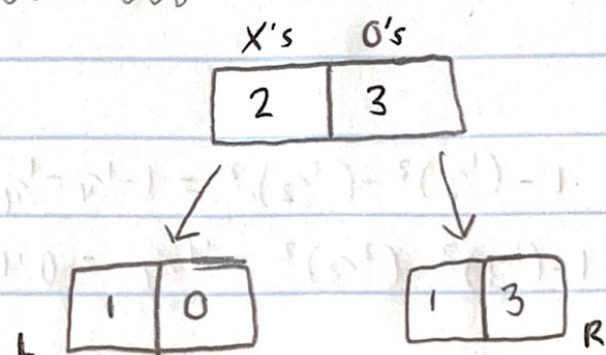


Problem 1) Decide the best vertical split of decision tree for the following dataset. Impurity can be measured by entropy or Gini-Index

For Split 1: $N=5$, $N_{\text{left}}=1$, $N_{\text{right}}=4$

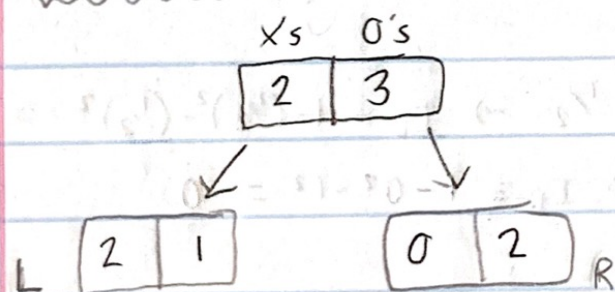


Left node $\rightarrow L: p_0 = 1/1 = 1$, $p_1 = 0/1 = 0 \rightarrow I_L = 1 - 1^2 - 0^2 = 0$

Right node $\rightarrow R: p_0 = 1/4$, $p_1 = 3/4 \rightarrow I_R = 1 - (1/4)^2 - (3/4)^2 = 6/16 = 0.375$

$$\text{Impurity of Split 1} = \frac{N_{\text{left}} \cdot I_L}{N} + \frac{N_{\text{right}} \cdot I_R}{N} = \frac{1}{5}(0) + \frac{4}{5}(0.375) = 0.30$$

For Split 3: $N=5$, $N_{\text{left}}=3$, $N_{\text{right}}=2$



Left node $\rightarrow L: p_0 = 2/3$, $p_1 = 1/3$, $I_L = 1 - (2/3)^2 - (1/3)^2 = 4/9 = 0.4444$

Right node $\rightarrow R: p_0 = 0/2 = 0$, $p_1 = 2/2 = 1$, $I_R = 1 - (0)^2 - (1)^2 = 0$

$$\text{Impurity of Split 3} = \frac{N_{\text{left}} \cdot I_L}{N} + \frac{N_{\text{right}} \cdot I_R}{N} = \frac{3}{5}(0.4444) + \frac{2}{5}(0) = 0.2667$$

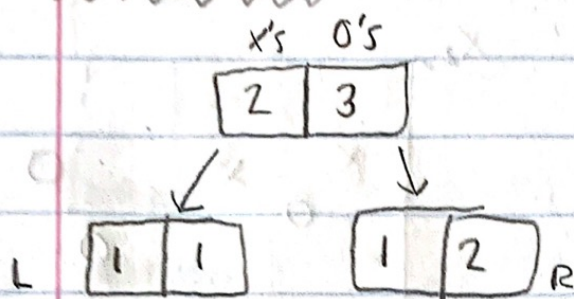
For Split 2:

1 0.30

2 0.2667

It has a lower impurity than split 1

For Split 2 : $N=5$, $N_{left}=2$, $N_{right}=3$

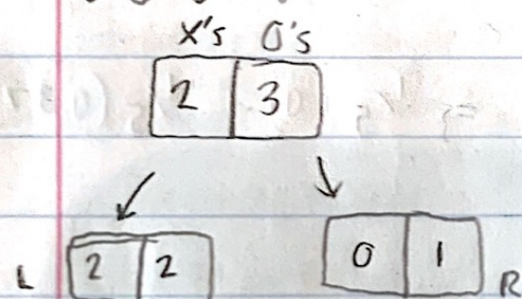


Left node $\rightarrow L$: $P_0 = 1/2$, $P_1 = 1/2 \rightarrow I_L = 1 - (1/2)^2 - (1/2)^2 = 1 - 1/4 - 1/4 = 0.5$

Right node $\rightarrow R$: $P_0 = 1/3$, $P_1 = 2/3 \rightarrow I_R = 1 - (1/3)^2 - (2/3)^2 = 4/9 = 0.4444$

Impurity of Split 3 = $\frac{N_{left}}{N} \cdot I_L + \frac{N_{right}}{N} \cdot I_R = \frac{2}{5}(0.50) + \frac{3}{5}(0.4444) = 0.40$

For Split 4 : $N=5$, $N_{left}=4$, $N_{right}=1$



Left Node $\rightarrow L$: $P_0 = 2/4 = 1/2$, $P_1 = 2/4 = 1/2 \rightarrow I_L = 1 - (1/2)^2 - (1/2)^2 = 0.50$

Right Node $\rightarrow R$: $P_0 = 0/1 = 0$, $P_1 = 1/1 = 1 \rightarrow I_R = 1 - 0^2 - 1^2 = 0$

Impurity of Split 4 = $\frac{N_{left}}{N} \cdot I_L + \frac{N_{right}}{N} \cdot I_R = \frac{4}{5}(0.50) + \frac{1}{5}(0) = 0.40$

Split	Impurity
1	0.30
2	0.4664
3	0.2667
4	0.40

By Gini Index, Split 3 is the best split because it has the lowest impurity measure of 0.2667

* Note that the calculations for Split 2 and Split 3 are out of order *

Problem 2)

Training data:

class	Sex	Survived
1	0	1
1	0	1
1	0	1
1	1	1
1	0	1
1	0	1
1	0	1
3	1	1
3	0	1
2	1	0
3	0	1
2	1	0
2	1	0
3	1	0
2	1	0
3	0	0
1	0	0
2	0	0
1	1	0
2	1	0

→ Use Gini Index as an impurity measure to:

- 1) Grow the maximum tree with three leaves (a stopping rule) of the data. Draw the diagram of the tree
- 2) Find the misclassification rate on training data of the maximal tree
- 3) Draw all possible subtrees
- 4) Validate the maximal tree and all the subtrees on the following data to select the optimal tree. Note: if the chance of survived is $\frac{1}{2}$, predict survived.

class	Sex	Survived
1	0	1
1	1	1
3	1	0
2	0	0

Problem 2 continued

Step 1: Decide the First Split

Split 1: Split sex into male & female

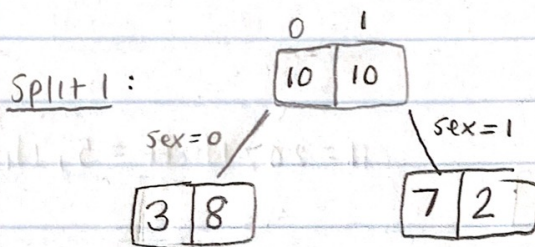
Split 2: Split class into class 1 and class 2,3

Split 3: Split class into class 2 and class 1,3

Split 4: Split class into class 3 and class 1,2

impurity of parent node for all splits $\rightarrow 10/20 = \text{Survived} \ \& \ 10/20 = \text{Not Survived}$

$$\text{impurity} = 1 - \left(\frac{10}{20}\right)^2 - \left(\frac{10}{20}\right)^2 = \boxed{0.50}$$



$$N=20, N_{\text{Left}}=11, N_{\text{Right}}=9$$

$$\text{left node: } p_0 = 3/11, p_1 = 8/11 \rightarrow I_L = 1 - \left(\frac{3}{11}\right)^2 - \left(\frac{8}{11}\right)^2$$

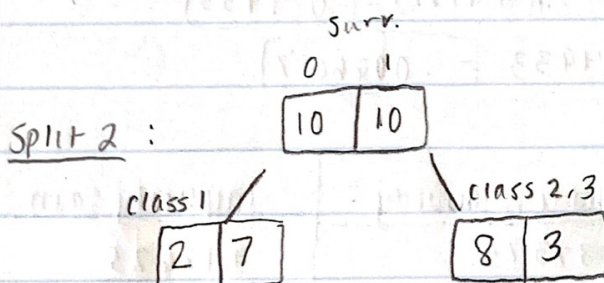
$$I_L = 0.3967$$

$$\text{right node: } p_0 = 7/9, p_1 = 2/9, I_R = 1 - \left(\frac{7}{9}\right)^2 - \left(\frac{2}{9}\right)^2$$

$$I_R = 0.3457$$

$$\text{impurity of split 1} = \frac{11}{20}(0.3967) + \frac{9}{20}(0.3457) = \boxed{0.37375}$$

$$\text{impurity gain of split 1} = 0.50 - 0.37375 = \boxed{0.12625}$$



$$N=20, N_{\text{Left}}=9, N_{\text{Right}}=11$$

$$\text{left node: } p_0 = 2/9, p_1 = 7/9 \rightarrow I_L = 1 - \left(\frac{2}{9}\right)^2 - \left(\frac{7}{9}\right)^2 = 0.3457$$

$$\text{right node: } p_0 = 8/11, p_1 = 3/11 \rightarrow I_R = 1 - \left(\frac{8}{11}\right)^2 - \left(\frac{3}{11}\right)^2 = 0.3967$$

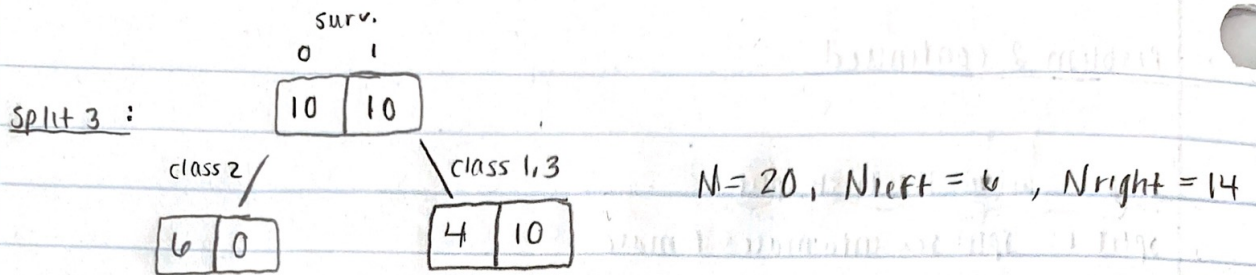
$$\text{impurity of split 2} = \frac{9}{20}(0.3457) + \frac{11}{20}(0.3967) = \boxed{0.37375}$$

$$\text{impurity gain of split 2} = 0.50 - 0.37375 = \boxed{0.12625}$$

Take cardines

MATH 460

Exam 1

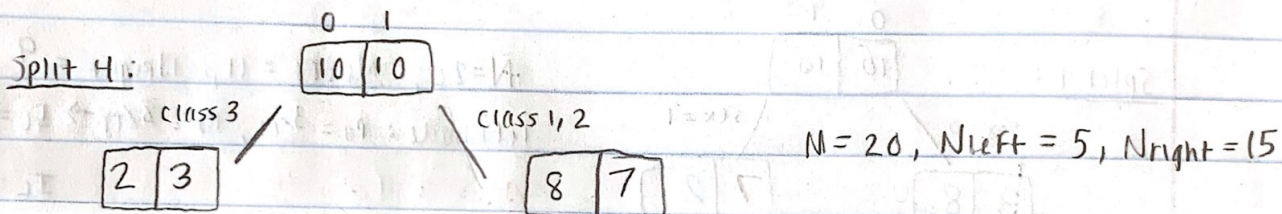


left node : $P_0 = 6/6 = 1, P_1 = 0/6 = 0 \rightarrow I_L = 1 - (1)^2 - (0)^2 = 0$

right node : $P_0 = 4/14, P_1 = 10/14 \rightarrow I_R = 1 - (4/14)^2 - (10/14)^2 = 0.40816$

impurity of Split 3 = $6/20(0) + 14/20(0.40816) = 0.28571$

impurity gain of Split 3 = $0.50 - 0.28571 = 0.2143$



left node : $P_0 = 2/5, P_1 = 3/5 \rightarrow I_L = 1 - (2/5)^2 - (3/5)^2 = 0.48$

right node : $P_0 = 8/15, P_1 = 7/15 \rightarrow I_R = 1 - (8/15)^2 - (7/15)^2 = 0.4978$

impurity of Split 4 = $5/20(0.48) + 15/20(0.4978) = 0.4933$

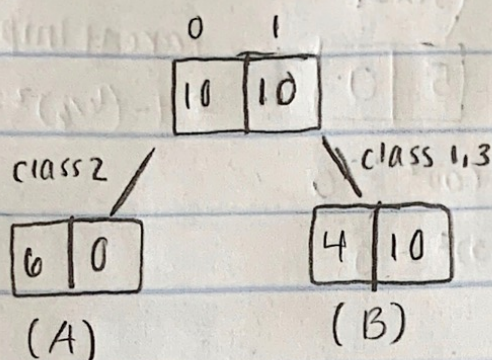
impurity gain of Split 4 = $0.50 - 0.4933 = 0.00667$

Split	Parent Impurity	Children Impurity	Impurity Gain
1	0.50	0.37375	0.12625
2	0.50	0.37375	0.12625
3	0.50	0.28571	0.2143
4	0.50	0.4933	0.00667

→ The best split is Split 3 since it has the highest impurity gain, so split 3 is the first of the tree.

Problem 2 - continued

using split 3, we have the first split as follows:



Step 2: Decide the second split → could happen at node A or node B

Looking at data in node A (class = 2):

class	sex	survived
2	1	0
2	1	0
2	1	0
2	1	0
2	0	0
2	1	0

candidate splits at node A? only candidate

is splitting at sex

Looking at data in node B (class = 1, 3):

class	sex	survived
-------	-----	----------

1	0	1
1	0	1
1	0	1
1	1	1
0	0	1
0	0	1
0	0	1
3	1	1
3	0	1
3	0	1
3	1	0
3	0	0
1	0	0
1	1	0

Node B can be split into

sex and class, so the candidates

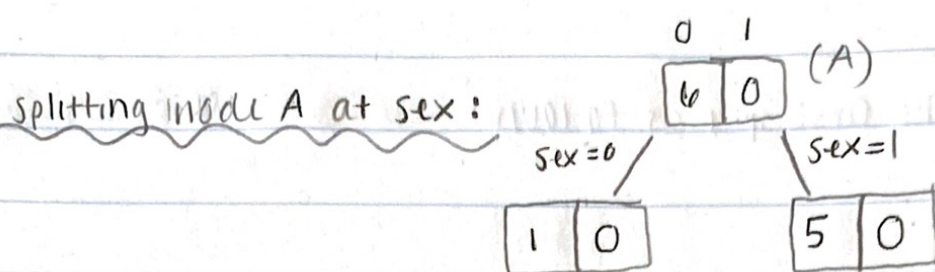
for the ^{2nd} split are:

- split node A at sex

- split node B at sex

- split node B at class

→ compute impurity of all candidate splits



$$N=6, N_{\text{left}}=1, N_{\text{right}}=5$$

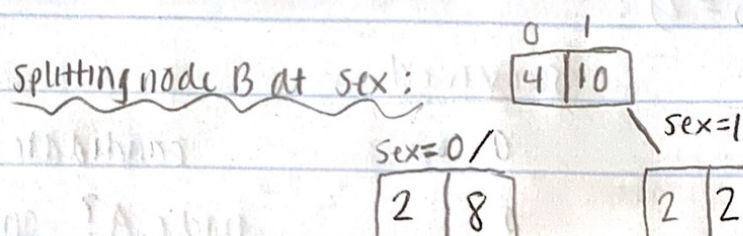
$$\text{Parent Impurity} = 1 - (1/6)^2 - (5/6)^2 = 0$$

$$\text{left node: } P_0 = 1/1, P_1 = 0/1 \rightarrow I_L = 1 - (1)^2 - (0)^2 = 0$$

$$\text{right node: } P_0 = 5/5, P_1 = 0/5 \rightarrow I_R = 1 - (1)^2 - (0)^2 = 0$$

$$\text{impurity of split} = 1/6(0) + 5/6(0) = 0$$

$$\text{impurity gain of split} = 0 - 0 = 0$$



$$N=14, N_{\text{left}}=10, N_{\text{right}}=4$$

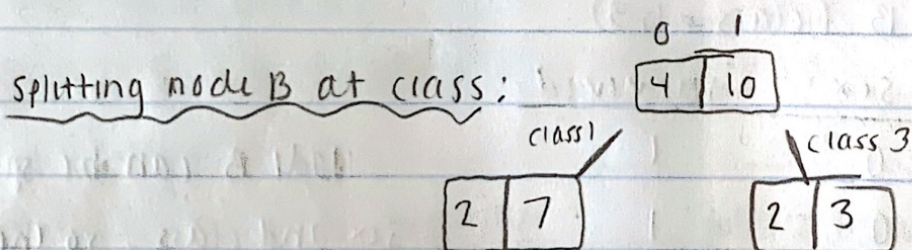
$$\text{left node: } P_0 = 2/10, P_1 = 8/10 \rightarrow I_L = 1 - (2/10)^2 - (8/10)^2 = 0.32$$

$$\text{right node: } P_0 = 2/4, P_1 = 2/4 \rightarrow I_R = 1 - (2/4)^2 - (2/4)^2 = 0.50$$

$$\text{impurity of split} = 10/14(0.32) + 4/14(0.50) = 0.3714$$

$$\text{parent impurity} = 1 - (4/14)^2 - (10/14)^2 = 0.40816$$

$$\text{impurity gain} = 0.40816 - 0.3714 = 0.03673$$



$$N=14, N_{\text{left}}=9, N_{\text{right}}=5$$

$$I_{\text{parent}} = 1 - (4/14)^2 - (10/14)^2 = 0.40816$$

$$\text{left node: } P_0 = 2/9, P_1 = 7/9, I_L = 1 - (2/9)^2 - (7/9)^2 = 0.3457$$

$$\text{right node: } P_0 = 2/5, P_1 = 3/5, I_R = 1 - (2/5)^2 - (3/5)^2 = 0.48$$

$$\text{impurity of split} = 9/14(0.3457) + 5/14(0.48) = 0.39366$$

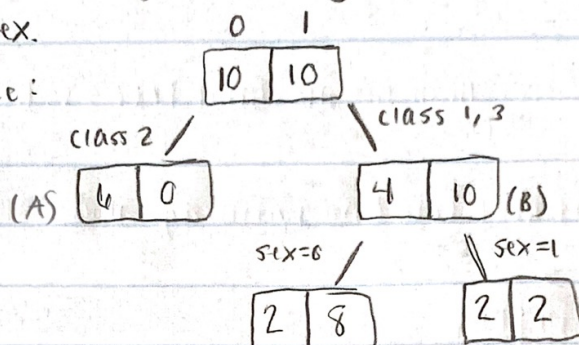
$$\text{impurity gain} = 0.40816 - 0.39366 = 0.0145$$

candidate split	parent impurity	children impurity	impurity gain
split A at sex	0	0	0
split B at sex	0.40816	0.4486	0.03673
split B at class	0.40816	0.39366	0.0145

Problem 2 - continued.

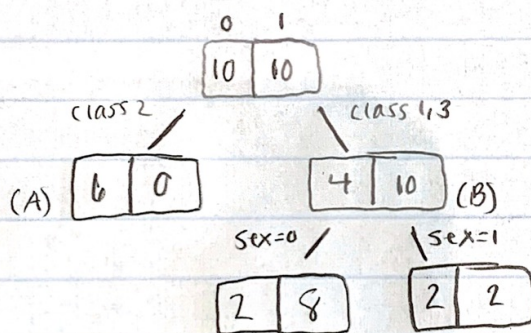
The best split has the highest impurity gain, so the second split is splitting node B at sex.

updated tree:



→ we now have 3 leaves

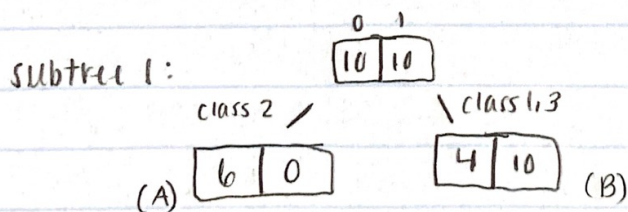
misclassification Rate: the prediction of each leaf is the majority. Thus, the misclassification rate of a leaf is the minority in the leaf



$$\text{Total misclassification} = 0 + 2 + 2 = 4$$

$$\text{misclassification rate} = 4 / (6 + 10 + 4) = 4/20 = 20\%$$

subtrees → only one possible



optimal tree

class	sex	Survived	Subtree 1 Predicts	Maximal Tree Predicts
1	0	1	1	1
1	1	1	1	1
3	1	0	1 (missed)	1 (missed)
2	0	0	0	0

→ Both the maximal tree and subtree 1 have one instance of misclassification, so we look to the simpler tree

→ Subtree 1 has less leaves than the maximal tree, so it's the simplest

→ Subtree 1 is the optimal tree and winning model

